

Yuheng Bu

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Department of Computer Science
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Homepage: <https://buyuheng.github.io/>

RESEARCH INTERESTS

My research interests lie in the intersection of *machine learning* and *information theory*. I have recently leveraged tools from information theory to develop *theoretically justified* learning algorithms for diverse applications, including model compression, anomaly detection, fair selective prediction, and uncertainty quantification. More broadly, the primary goal of my research is to lay information-theoretic foundations for trustworthy learning algorithms, particularly with generalization, fairness, and privacy guarantees.

ACADEMIC APPOINTMENTS

- Assistant Professor, Department of Computer Science,
College of Engineering, **University of California, Santa Barbara** Jul. 2025–present
- Assistant Professor, Department of Electrical and Computer Engineering,
Herbert Wertheim College of Engineering, **University of Florida** Aug. 2022–Jun. 2025
- Postdoctoral Research Associate, **Massachusetts Institute of Technology** Sep. 2019–May 2022
Advisor: Gregory W. Wornell
- Undergraduate Visiting Research (UGVR), **Stanford University** Jun. 2013–Sep. 2013
Only 18 students chosen from mainland China and Taiwan
Advisor: Tsachy Weissman

EDUCATION

- Ph.D. in Electrical and Computer Engineering
University of Illinois at Urbana-Champaign Jan. 2017–Aug. 2019
Advisor: Venugopal V. Veeravalli
Thesis: “Information-theoretic Bounds in Learning Algorithms”
- Master in Electrical and Computer Engineering
University of Illinois at Urbana-Champaign Aug. 2014–Dec. 2016
Advisor: Venugopal V. Veeravalli
Thesis: “Estimation of KL Divergence: Optimal Minimax Rate”
- B.E. (**with honors**) in Electronic Engineering
Tsinghua University, Beijing, China Aug. 2010–Jul. 2014
Double Major in **Economics**

SPONSORED RESEARCH

- **CAREER**: LLM Watermarking and Beyond: Foundations and Algorithms via Distributional Information Embedding
PI, 04/15/2026–03/31/2031, **NSF**
- EPCN: VISUALS: Verifiable Information-Theoretic Safety Under Augmented Latent Shifts
Co-PI, in collaboration with Ivan Ruchkin (PI, UF), Jose Principe (Co-PI, UF)
10/01/2025–09/30/2028, **NSF**

- Collaborative Research: Elements: Empowering Semiconductor Device Research and Education through Integrated Machine Learning Models and Database
Co-PI, in collaboration with Jing Guo (PI, UF), Erik Deumens (Co-PI, UF)
08/15/2024–07/31/2027, **NSF**
- UF Research Opportunity Seed Fund (ROSF) 2023: Developing Graph Neural Networks to Quantify Uncertainty for Future of Remote Work in Cities.
Co-PI, in collaboration with Shenhao Wang (PI, UF), Lingqian Hu (Co-PI, UF)
09/01/2023–08/31/2025, **UF**

FULL PUBLICATION LIST

Journal Papers

- [1] Z. Cheng, L. Hu, **Y. Bu**, Y. Zhou, S. Wang, “Graph neural networks for residential location choice: Connection to classical logit models,” *Transportation Research Part B: Methodological*, vol. 209, p. 103 464, 2026. DOI: [10.1016/j.trb.2026.103464](https://doi.org/10.1016/j.trb.2026.103464).
- [2] F. Laakom, M. Gabbouj, J. Schmidhuber, **Y. Bu**, “Class-wise generalization error: An information-theoretic analysis,” *Transactions on Machine Learning Research*, 2025. [Online]. Available: <https://openreview.net/forum?id=asW4VcDFpi>.
- [3] G. Aminian*, **Y. Bu***, L. Toni, M. R. Rodrigues, G. W. Wornell, “Information-theoretic characterizations of generalization error for the Gibbs algorithm,” *IEEE Transactions on Information Theory*, vol. 70, no. 1, pp. 632–655, 2024. DOI: [10.1109/TIT.2023.3329617](https://doi.org/10.1109/TIT.2023.3329617) (* equal contribution).
- [4] J. K. Lee*, **Y. Bu***, P. Sattigeri, R. Panda, G. W. Wornell, L. Karlinsky, R. S. Feris, “A maximal correlation framework for fair machine learning,” *Entropy*, vol. 24, no. 4, p. 461, 2022. DOI: [10.3390/e24040461](https://doi.org/10.3390/e24040461) (* equal contribution).
- [5] **Y. Bu**, W. Gao, S. Zou, V. V. Veeravalli, “Population risk improvement with model compression: An information-theoretic approach,” *Entropy*, vol. 23, no. 10, p. 1255, 2021. DOI: [10.3390/e23101255](https://doi.org/10.3390/e23101255).
- [6] **Y. Bu**, S. Zou, V. V. Veeravalli, “Tightening mutual information based bounds on generalization error,” *IEEE Journal on Selected Areas in Information Theory*, vol. 1, no. 1, pp. 121–130, 2020. DOI: [10.1109/JSAIT.2020.2991139](https://doi.org/10.1109/JSAIT.2020.2991139).
- [7] C. Wilson, **Y. Bu**, V. V. Veeravalli, “Adaptive sequential machine learning,” *Sequential Analysis*, vol. 38, no. 4, pp. 545–568, 2019. DOI: [10.1080/07474946.2019.1686889](https://doi.org/10.1080/07474946.2019.1686889) (16th Abraham Wald Prize).
- [8] **Y. Bu**, S. Zou, V. V. Veeravalli, “Linear-complexity exponentially-consistent tests for universal outlying sequence detection,” *IEEE Transactions on Signal Processing*, vol. 67, no. 8, pp. 2115–2128, 2019. DOI: [10.1109/TSP.2019.2901364](https://doi.org/10.1109/TSP.2019.2901364).
- [9] **Y. Bu***, S. Zou*, Y. Liang, V. V. Veeravalli, “Estimation of KL divergence: Optimal minimax rate,” *IEEE Transactions on Information Theory*, vol. 64, no. 4, pp. 2648–2674, 2018. DOI: [10.1109/TIT.2018.2805844](https://doi.org/10.1109/TIT.2018.2805844) (* equal contribution).

Conference and Workshop Publications

- [1] Y. Liu, Y. Huang, Y.-X. Wang, Y. Liang, **Y. Bu**, “ConvexBench: Can LLMs recognize convex functions?” In *Proc. International Conference on Machine Learning (ICML)*, 2026 (**acceptance rate: 27%**).
- [2] Y. Liu, X. Zhao, D. Song, G. W. Wornell, **Y. Bu**, “Position: LLM watermarking should align stakeholders’ incentives for practical adoption,” in *Findings of the Association for Computational Linguistics: ACL 2026*, 2026, pp. 25 887–25 903 (**acceptance rate: 37%**).
- [3] Y. Liu, X. Zhao, C. Kruegel, D. Song, **Y. Bu**, “In-context watermarks for large language models,” in *Proc. International Conference on Learning Representations (ICLR)*, 2026. [Online]. Available: <https://openreview.net/forum?id=fD9YRHazW3> (**acceptance rate: 28%**).

- [4] L. An, Y. Liu, Y. Liu, **Y. Bu**, Y. Zhang, S. Chang, “A reinforcement learning framework for robust and secure LLM watermarking,” in *Proc. European Chapter of the Association for Computational Linguistics (EACL)*, 2026, pp. 7181–7198. DOI: [10.18653/v1/2026.eacl-long.338](https://doi.org/10.18653/v1/2026.eacl-long.338) (acceptance rate: 20%).
- [5] D. Yu, R. Xu, D. Zhuang, **Y. Bu**, S. Wang, G. Wang, “TrustEnergy: A unified framework for accurate and reliable user-level energy usage prediction,” in *Proc. AAAI Conference on Artificial Intelligence (AAAI)*, Special Track on AI for Social Impact, 2026, pp. 39 558–39 566. DOI: [10.1609/aaai.v40i46.41307](https://doi.org/10.1609/aaai.v40i46.41307) (acceptance rate: 24%).
- [6] H. He, Y. Liu, Z. Wang, Y. Mao, **Y. Bu**, “Theoretically grounded framework for LLM watermarking: A distribution-adaptive approach,” in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2025. [Online]. Available: https://papers.nips.cc/paper_files/paper/2025/hash/956f80e60a211e3a0f6ec5f2dfa3551a-Abstract-Conference.html (acceptance rate: 25%).
- [7] D. Yu, D. Zhuang, L. Jiang, R. Xu, X. Ye, **Y. Bu**, S. Wang, G. Wang, “UQGN: Uncertainty quantification of graph neural networks for multivariate spatiotemporal prediction,” in *Proc. ACM International Conference on Advances in Geographic Information Systems*, 2025, pp. 52–65. DOI: [10.1145/3748636.3762709](https://doi.org/10.1145/3748636.3762709).
- [8] H. He, Y. Liu, Z. Wang, Y. Mao, **Y. Bu**, “Distributional information embedding: A framework for multi-bit watermarking,” in *Asia Pacific Workshop on Data Science and Information Theory (APWDSIT)*, 2025. DOI: [10.1109/APWDSIT64675.2025.11478544](https://doi.org/10.1109/APWDSIT64675.2025.11478544).
- [9] L. An, Y. Liu, Y. Liu, Y. Zhang, **Y. Bu**, S. Chang, “Defending LLM watermarking against spoofing attacks with contrastive representation learning,” in *Proc. Conference on Language Modeling (COLM)*, 2025. [Online]. Available: <https://openreview.net/forum?id=n5hmtkd17k> (acceptance rate: 32%).
- [10] F. Laakom, H. Chen, J. Schmidhuber, **Y. Bu**, “Fairness overfitting in machine learning: An information-theoretic perspective,” in *Proc. International Conference on Machine Learning (ICML)*, 2025. [Online]. Available: <https://proceedings.mlr.press/v267/laakom25a.html> (acceptance rate: 29%).
- [11] Y. Liu, Y. Song, H. Ci, Y. Zhang, H. Wang, M. Z. Shou, **Y. Bu**, “Image watermarks are removable using controllable regeneration from clean noise,” in *Proc. International Conference on Learning Representations (ICLR)*, 2025. [Online]. Available: <https://openreview.net/forum?id=mDKx1fraAn> (acceptance rate: 32%).
- [12] J. J. Ryu, M. Shen, S. Ghosh, **Y. Bu**, P. Sattigeri, S. Das, G. W. Wornell, “Are uncertainty quantification capabilities of evidential deep learning a mirage?” In *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, 2024. DOI: [10.52202/079017-3424](https://doi.org/10.52202/079017-3424) (acceptance rate: 26%).
- [13] Y. Zhu, **Y. Bu**, “Information-theoretic analysis of the Gibbs algorithm: An individual sample approach,” in *2024 IEEE Information Theory Workshop (ITW)*, IEEE, 2024, pp. 537–542. DOI: [10.1109/ITW61385.2024.10806933](https://doi.org/10.1109/ITW61385.2024.10806933).
- [14] D. Zhuang, **Y. Bu**, G. Wang, S. Wang, J. Zhao, “SAUC: Sparsity-aware uncertainty calibration for spatiotemporal prediction with graph neural networks,” in *Proc. ACM International Conference on Advances in Geographic Information Systems*, 2024, pp. 160–172. DOI: [10.1145/3678717.3691241](https://doi.org/10.1145/3678717.3691241).
- [15] B. Hu, **Y. Bu**, J. C. Príncipe, “Learning orthonormal features in self-supervised learning using functional maximal correlation,” in *2024 IEEE International Conference on Image Processing (ICIP)*, IEEE, 2024, pp. 472–478. DOI: [10.1109/ICIP51287.2024.10648197](https://doi.org/10.1109/ICIP51287.2024.10648197).
- [16] H. Chen, G. Aminian, **Y. Bu**, “An algorithm for computing the capacity of symmetrized KL information for discrete channels,” in *2024 60th Annual Allerton Conference on Communication, Control, and Computing*, IEEE, 2024, pp. 1–8. DOI: [10.1109/Allerton63246.2024.10735330](https://doi.org/10.1109/Allerton63246.2024.10735330).
- [17] C. Shi, **Y. Bu**, J. Fu, “Information-theoretic opacity-enforcement in markov decision processes,” in *Proc. International Joint Conference on Artificial Intelligence (IJCAI)*, 2024, pp. 6779–6787. DOI: [10.24963/ijcai.2024/749](https://doi.org/10.24963/ijcai.2024/749) (acceptance rate: 15%).

- [18] Y. Liu, **Y. Bu**, “Adaptive text watermark for large language models,” in *Proc. International Conference on Machine Learning (ICML)*, PMLR, 2024, pp. 30 718–30 737. [Online]. Available: <https://proceedings.mlr.press/v235/liu24e.html> (acceptance rate: 28%).
- [19] J. J. Ryu, X. Xu, H. S. M. Erol, **Y. Bu**, L. Zheng, G. W. Wornell, “Operator SVD with neural networks via nested low-rank approximation,” in *Proc. International Conference on Machine Learning (ICML)*, PMLR, 2024, pp. 42 870–42 905. [Online]. Available: <https://proceedings.mlr.press/v235/ryu24b.html> (acceptance rate: 28%).
- [20] **Y. Bu**, “Towards optimal inverse temperature in the Gibbs algorithm,” in *2024 IEEE International Symposium on Information Theory (ISIT)*, IEEE, 2024, pp. 2257–2262. DOI: [10.1109/ISIT57864.2024.10619441](https://doi.org/10.1109/ISIT57864.2024.10619441).
- [21] A. Shah, M. Shen, J. J. Ryu, S. Das, P. Sattigeri, **Y. Bu**, G. W. Wornell, “Group fairness with uncertain sensitive attributes,” in *2024 IEEE International Symposium on Information Theory (ISIT)*, IEEE, 2024, pp. 208–213. DOI: [10.1109/ISIT57864.2024.10619400](https://doi.org/10.1109/ISIT57864.2024.10619400).
- [22] H. Chen, G. W. Wornell, **Y. Bu**, “Gibbs-based information criteria and the over-parameterized regime,” in *Proc. International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, 2024, pp. 4501–4509. [Online]. Available: <https://proceedings.mlr.press/v238/chen24g.html> (acceptance rate: 28%).
- [23] M. Shen, **Y. Bu**, G. W. Wornell, “On balancing bias and variance in unsupervised multi-source-free domain adaptation,” in *Proc. International Conference on Machine Learning (ICML)*, PMLR, 2023, pp. 30 976–30 991. [Online]. Available: <https://proceedings.mlr.press/v202/shen23b.html> (acceptance rate: 28%).
- [24] **Y. Bu**, H. V. Tetali, G. Aminian, M. R. Rodrigues, G. W. Wornell, “On the generalization error of Meta learning for the Gibbs algorithm,” in *Proc. IEEE International Symposium on Information Theory (ISIT)*, IEEE, 2023, pp. 2488–2493. DOI: [10.1109/ISIT54713.2023.10206566](https://doi.org/10.1109/ISIT54713.2023.10206566).
- [25] A. Weiss, A. Lancho, **Y. Bu**, G. W. Wornell, “A bilateral bound on the mean-square error for estimation in model mismatch,” in *Proc. IEEE International Symposium on Information Theory (ISIT)*, IEEE, 2023, pp. 2655–2660. DOI: [10.1109/ISIT54713.2023.10206620](https://doi.org/10.1109/ISIT54713.2023.10206620).
- [26] M. Shen, S. S. Ghosh, P. Sattigeri, S. Das, **Y. Bu**, G. W. Wornell, “Reliable gradient-free and likelihood-free prompt tuning,” in *Findings of the Association for Computational Linguistics: EACL 2023*, 2023, pp. 2416–2429. DOI: [10.18653/v1/2023.findings-eacl.183](https://doi.org/10.18653/v1/2023.findings-eacl.183).
- [27] H. He, G. Aminian, **Y. Bu**, M. R. Rodrigues, V. Y. Tan, “How does pseudo-labeling affect the generalization error of the semi-supervised Gibbs algorithm?” In *Proc. International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, 2023, pp. 8494–8520. [Online]. Available: <https://proceedings.mlr.press/v206/he23b.html> (acceptance rate: 29%).
- [28] M. Shen, **Y. Bu**, P. Sattigeri, S. S. Ghosh, S. Das, G. W. Wornell, “Post-hoc uncertainty learning using a Dirichlet Meta-model,” in *Proc. AAAI Conference on Artificial Intelligence (AAAI)*, 2023. DOI: [10.1609/aaai.v37i8.26167](https://doi.org/10.1609/aaai.v37i8.26167) (acceptance rate: 19%).
- [29] A. Lancho, A. Weiss, G. C. Lee, J. Tang, **Y. Bu**, Y. Polyanskiy, G. W. Wornell, “Data-driven blind synchronization and interference rejection for digital communication signals,” in *IEEE Global Communications Conference (GLOBECOM)*, IEEE, 2022, pp. 2296–2302. DOI: [10.1109/GLOBECOM48099.2022.10001513](https://doi.org/10.1109/GLOBECOM48099.2022.10001513).
- [30] G. C. Lee, A. Weiss, A. Lancho, J. Tang, **Y. Bu**, Y. Polyanskiy, G. W. Wornell, “Exploiting temporal structures of cyclostationary signals for data-driven single-channel source separation,” in *Proc. IEEE International Workshop on Machine Learning for Signal Processing (MLSP)*, IEEE, 2022, pp. 1–6. DOI: [10.1109/MLSP55214.2022.9943311](https://doi.org/10.1109/MLSP55214.2022.9943311) (Best Student Paper Award).
- [31] A. Shah*, **Y. Bu***, J. K. Lee, P. Sattigeri, R. Panda, S. Das, G. W. Wornell, “Selective regression under fairness criteria,” in *Proc. International Conference on Machine Learning (ICML)*, PMLR, 2022, pp. 19 598–19 615. [Online]. Available: <https://proceedings.mlr.press/v162/shah22a.html> (* equal contribution; acceptance rate: 22%).

- [32] G. Aminian*, **Y. Bu***, G. W. Wornell, M. R. Rodrigues, “Tighter expected generalization error bounds via convexity of information measures,” in *Proc. IEEE International Symposium on Information Theory (ISIT)*, IEEE, 2022, pp. 2481–2486. DOI: [10.1109/ISIT50566.2022.9834474](https://doi.org/10.1109/ISIT50566.2022.9834474) (* equal contribution).
- [33] J. K. Lee*, **Y. Bu***, P. Sattigeri, R. Panda, G. W. Wornell, L. Karlinsky, R. S. Feris, “A maximal correlation approach to imposing fairness in machine learning,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2022, pp. 3523–3527. DOI: [10.1109/ICASSP43922.2022.9747263](https://doi.org/10.1109/ICASSP43922.2022.9747263) (* equal contribution).
- [34] **Y. Bu***, G. Aminian*, L. Toni, G. W. Wornell, M. R. Rodrigues, “Characterizing and understanding the generalization error of transfer learning with Gibbs algorithm,” in *Proc. International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR, 2022, pp. 8673–8699. [Online]. Available: <https://proceedings.mlr.press/v151/bu22a.html> (* equal contribution; **acceptance rate: 29%**).
- [35] G. Aminian*, **Y. Bu***, L. Toni, M. R. Rodrigues, G. W. Wornell, “An exact characterization of the generalization error for the Gibbs algorithm,” in *Proc. Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8106–8118. [Online]. Available: <https://papers.nips.cc/paper/2021/hash/445e24b5f22cacb9d51a837c10e91a3f-Abstract.html> (* equal contribution; **acceptance rate: 26%**).
- [36] J. K. Lee*, **Y. Bu***, D. Rajan, P. Sattigeri, R. Panda, S. Das, G. W. Wornell, “Fair selective classification via sufficiency,” in *Proc. International Conference on Machine Learning (ICML)*, PMLR, 2021, pp. 6076–6086. [Online]. Available: <https://proceedings.mlr.press/v139/lee21b.html> (* equal contribution; **Oral; Top 3%**).
- [37] **Y. Bu**, T. Wang, G. W. Wornell, “SDP methods for sensitivity-constrained privacy funnel and information bottleneck problems,” in *Proc. IEEE International Symposium on Information Theory (ISIT)*, IEEE, 2021, pp. 49–54. DOI: [10.1109/ISIT45174.2021.9517714](https://doi.org/10.1109/ISIT45174.2021.9517714).
- [38] **Y. Bu**, W. Gao, S. Zou, V. V. Veeravalli, “Information-theoretic understanding of population risk improvement with model compression,” in *Proc. AAAI Conference on Artificial Intelligence (AAAI)*, vol. 34, 2020, pp. 3300–3307. DOI: [10.1609/aaai.v34i04.5730](https://doi.org/10.1609/aaai.v34i04.5730).
- [39] **Y. Bu**, K. Small, “Active learning in recommendation systems with multi-level user preferences,” in *AAAI Workshop on Interactive and Conversational Recommendation Systems (WICRS)*, 2020.
- [40] **Y. Bu**, J. Lu, V. V. Veeravalli, “Active and adaptive sequential learning with per time-step excess risk guarantees,” in *Proc. IEEE Asilomar Conference on Signals, Systems, and Computers*, IEEE, 2019, pp. 1606–1610. DOI: [10.1109/IEEECONF44664.2019.9048932](https://doi.org/10.1109/IEEECONF44664.2019.9048932).
- [41] **Y. Bu**, S. Zou, V. V. Veeravalli, “Tightening mutual information based bounds on generalization error,” in *Proc. IEEE International Symposium on Information Theory (ISIT)*, IEEE, 2019, pp. 587–591. DOI: [10.1109/ISIT.2019.8849590](https://doi.org/10.1109/ISIT.2019.8849590).
- [42] **Y. Bu**, J. Lu, V. V. Veeravalli, “Model change detection with application to machine learning,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2019, pp. 5341–5346. DOI: [10.1109/ICASSP.2019.8682153](https://doi.org/10.1109/ICASSP.2019.8682153).
- [43] **Y. Bu**, S. Zou, V. V. Veeravalli, “Linear-complexity exponentially-consistent tests for universal outlying sequence detection,” in *Proc. IEEE International Symposium on Information Theory (ISIT)*, IEEE, 2017, pp. 988–992. DOI: [10.1109/ISIT.2017.8006676](https://doi.org/10.1109/ISIT.2017.8006676).
- [44] **Y. Bu***, S. Zou*, Y. Liang, V. V. Veeravalli, “Estimation of KL divergence between large-alphabet distributions,” in *Proc. IEEE International Symposium on Information Theory (ISIT)*, IEEE, 2016, pp. 1118–1122. DOI: [10.1109/ISIT.2016.7541473](https://doi.org/10.1109/ISIT.2016.7541473) (* equal contribution).
- [45] **Y. Bu**, S. Zou, Y. Liang, V. V. Veeravalli, “Universal outlying sequence detection for continuous observations,” in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2016, pp. 4254–4258. DOI: [10.1109/ICASSP.2016.7472479](https://doi.org/10.1109/ICASSP.2016.7472479).

Patents

- [1] M. Shen, **Y. Bu**, P. Sattigeri, S. Ghosh, S. Das, G. Wornell, *Post-hoc uncertainty quantification for machine learning systems*, US Patent App. 18/538,726, Jun. 2025.
- [2] M. Shen, S. Ghosh, P. Sattigeri, S. Das, **Y. Bu**, G. Wornell, *Reliable gradient-free and likelihood-free prompt tuning*, US Patent App. 18/510,612, May 2025.
- [3] A. Shah, M. Shen, S. Das, P. Sattigeri, **Y. Bu**, G. Wornell, *Sensitive attribute driven predictive modeling*, US Patent App. 18/503,166, May 2025.
- [4] J. K.-W. Lee, **Y. Bu**, D. Rajan, P. Sattigeri, S. Das, R. Panda, G. Wornell, *Fair selective classification via a variational mutual information upper bound for imposing sufficiency*, US Patent App. 17/565,411, Jun. 2023.

TUTORIALS

- “Characterizing the Generalization Error of Machine Learning Algorithms Via Information Measures,”
IEEE Information Theory Workshop (ITW) 2024

INVITED TALKS

- Adaptive Text Watermark for Large Language Models
 - Information Science and Learning Systems Seminar, Texas A&M Nov. 2024
 - Asilomar Conference on Signals, Systems, and Computers, Pacific Grove Oct. 2024
 - CS Colloquium, University of California, Santa Barbara (UCSB) Sep. 2024
 - Allerton Conference on Communication, Control, and Computing, UIUC Sep. 2024
 - Meta Fundamental AI Research (FAIR), Seattle Aug. 2024
 - Annual Conference on Information Sciences and Systems, Princeton University Mar. 2024
- Towards Consistent Uncertainty Quantification using Deep Models
 - 2024 Joint Statistical Meetings - Statistics & Data Science, Portland Aug. 2024
- Group Fairness with Uncertainty in Sensitive Attributes
 - Allerton Conference on Communication, Control, and Computing, UIUC Oct. 2023
- From Sensitivity-constrained Information Bottleneck to Fair Selective Prediction
 - Information Theory and Data Science Workshop (Virtual), NUS Jan. 2023
- Post-hoc Uncertainty Learning using a Dirichlet Meta-Model
 - Allerton Conference on Communication, Control, and Computing, UIUC Oct. 2022
- Can Information Theory Characterize Learning Algorithms?
 - ECE department, The Ohio State University (OSU) Mar. 2024
 - Information Theory and Applications Workshop (ITA), UCSD Feb. 2023
 - CS Colloquium, University of California, Santa Barbara (UCSB) Oct. 2022
 - ECE department, University of Florida (UF) Mar. 2022
 - ECE department, National University of Singapore (NUS) Jan. 2022
 - ECE department, Hong Kong University of Science and Technology (HKUST) Jan. 2022
 - EEE department, Hong Kong University (HKU) Dec. 2021

HONORS & AWARDS

- 16th **Abraham Wald Prize**, *Sequential Analysis* 2023
- **MLSP** 2022 Best Student Paper Award 2022
- Outstanding Reviewer Award, **NeurIPS** 2021
- **Yi-Min Wang and Pi-Yu Chung** Research Award, UIUC 2019
- Nominee for Graduation Day at **IEEE ITA Workshop** 2019
- Student Travel Grant, **IEEE ISIT** 2016, 2017
- Student Travel Grant, **IEEE ICASSP** 2016
- **Outstanding graduate**, Tsinghua University 2014
- **National Scholarship** Granted by Ministry of Education of China (**top 2%**) 2012, 2013

TEACHING EXPERIENCE

- **University of Florida:**
 - EEL 3850: Data Science for ECE Fall 2023, 2024
 - EEL 6935: Information-Theoretic Methods in Machine Learning Spring 2023, 2024, 2025
- Teaching and developing content for **MIT IDSS MicroMasters** online program:
 - 6.86x Machine Learning with Python–From Linear Models to Deep Learning Spring 2020
 - 14.310x Data Analysis for Social Scientists (live recitation and video recording) Fall 2019
- Teaching assistant at **UIUC**
 - ECE 365: Data Science and Engineering Spring 2019
 - ECE 398: Making Sense of Big Data Spring 2017, Fall 2018
 - ECE 598: Computational Inference and Learning Fall 2016

PROFESSIONAL ACTIVITIES

- **Panelist:** NSF CISE: CIF
- **Guest Editor:** Entropy Special Issue “Fairness in Machine Learning: Information Theoretic Perspectives,” 2022–2023.
- **Organizer:** 6th–8th Annual Workshop on Cognition & Control at University of Florida, 2023–2025
- **Reviewer:** Journal of Machine Learning Research (JMLR), Transactions on Machine Learning Research (TMLR), IEEE Transactions on Information Theory (TIT), IEEE Journal on Selected Areas in Information Theory (JSAIT), IEEE Transactions on Signal Processing (TSP), IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), IEEE Transactions on Neural Networks and Learning Systems (TNNLS), Transactions on Information Forensics & Security (TIFS), IEEE Transactions on Vehicular Technology (TVT), IEEE Journal on Selected Areas in Communications (JSAC) Conferences: NeurIPS, ICML, ICLR, AAAI, AISTATS, IJCAI, ISIT, ITW, ICASSP
- Area Chair: NeurIPS (2025), ICLR (2026), ICML (2026)
- Technical Program Committee (TPC) Member: ISIT (2025, 2026), ITW (2025)